



Investigating the Effect of Drag Reduction Agents on Heavy Crude Oil Flow in Pipelines: A Review

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Abstract: Considering the combination of escalating global energy request and the decrease in traditional petroleum resources, heavy crude oils are widely regarded as a prospective source of energy in the future. In numerous regions around the world, heavy crude must be carried through pipeline systems that connect the production fields to either storage terminals or refining plants. The conveyance of heavy crude oils necessitates the implementation of efficient pumping methodologies to reduce operational costs during the midstream phase. This objective might be achieved by diminishing both the viscosity of the oil and the frictional losses resulting from flow conditions. The concept of drag reduction in pipelines has garnered significant attention over the recent few decades owing to its prospective engineering implementations, particularly within industries involved in fluid transportation. Augmenting the flowing of crude with minute amounts of drag-reducing agents (DRAs) is capable of minimizing the decline in pressure across pipelines. Extensive surveys have been performed on DRA as a viable approach to alleviate the obstacles posed by increased resistance during oil transportation. DRAs such as surfactants, nanoparticles, bio-additives, polymers, and fibers are mixed with diluted crude oils to inhibit the formation of turbulent eddies, thereby facilitating higher flow rates under consistent pressure conditions. This research discusses the potential advantages of incorporating DRAs in heavy crude oil pipelines, including improved flow rates, reduced energy consumption, and prolonged pipeline lifespan. In essence, this review consolidates the current understanding of the influence of DRAs on the inflow of heavy crude oils in pipelines and highlights areas for future research to enhance the utilization of DRAs and tackle existing obstacles, ultimately contributing to a more effective and sustainable transportation of heavy crude oils.

Keywords: Additives; Drag reduction; Drag-reducing agents; Heavy crude oil; Pipelines

1 Introduction

Countries with acute fuel shortages need crude oil for transportation, power, and industry. Thus, crude oil is mostly used in energy-producing industries [1]. Economic expansion and population growth have boosted demand for fossil fuels, causing conventional oil stocks, notably light and medium crude oil, to decrease and become in short supply to meet energy demand [2]. Heavy oils are denser, viscous, and have more resins, asphaltenes, saturated compounds, and aromatics than light oils [3]. All of the following are examples of alternative fossil fuels: Heavy crude, Extra-Heavy crude, crude Shale, oil Sands, Tar Sands, and Bitumen. In contrast to conventional oil, which has relatively Low-cost production, natural non-conventional oil cannot be extracted and produced in the same manner. Because their complicated composition makes it difficult for them to flow freely without flow assurance difficulties [4]. Heavy oil reserves, both those that already exist and those that are likely to be discovered in the future, are anticipated to be estimated to be twice as large as conventional oil reserves [5]. In 2016, the China National Petroleum Corporation (CNPC) produced a report assessing global oil and gas resources. According to this analysis, there are 127 billion tons of unconventional oil that may be recovered worldwide [6]. While the Middle East is the owner of 14% of this reserve, the continents of North and South America combined control 57% of the total resources that are recoverable. Figure 1 shows that reserves can also be discovered in the regions of Asia, Russia, Europe, and Africa. It is anticipated that heavy oil will contribute a greater proportion of the world supply of liquid hydrocarbons as a result of developments in oil recovery technology and the diminishing availability of light oil [7].

Figure 2 illustrates the projected market shares of heavy crudes over the next 20 years. Because of the large presence of high molecular weight elements such as waxes, asphaltenes, and resins, the viscosity is enhanced, ranging from 103 to 106 cP, and the American Petroleum Institute gravity (API gravity, °API) is reduced, namely below 20 for heavy crude and 10 for extra heavy crude. These two characteristics are primarily the result of the excessive presence of these constituents. Furthermore, asphaltene is the strongly polar variant of polycyclic aromatic hydrocarbon since it contains heteroatoms and metals [8]. This is because asphaltene comprises both of these elements. This polarity causes the creation of a viscoelastic network of nanoaggregates, which ultimately produces an expansion in the viscosity of the material [3, 9]. In addition, when the temperature drops under the pour point, the heavy crude begins to solidify, which presents a considerable barrier for transportation, particularly in areas that are cold offshore. Due to wax and asphaltene buildup on the inside of pipelines, a substantial pressure in pipelines is reduced. Eventually this will minimize the efficient diameter of the flow and ultimately block the lines [10].

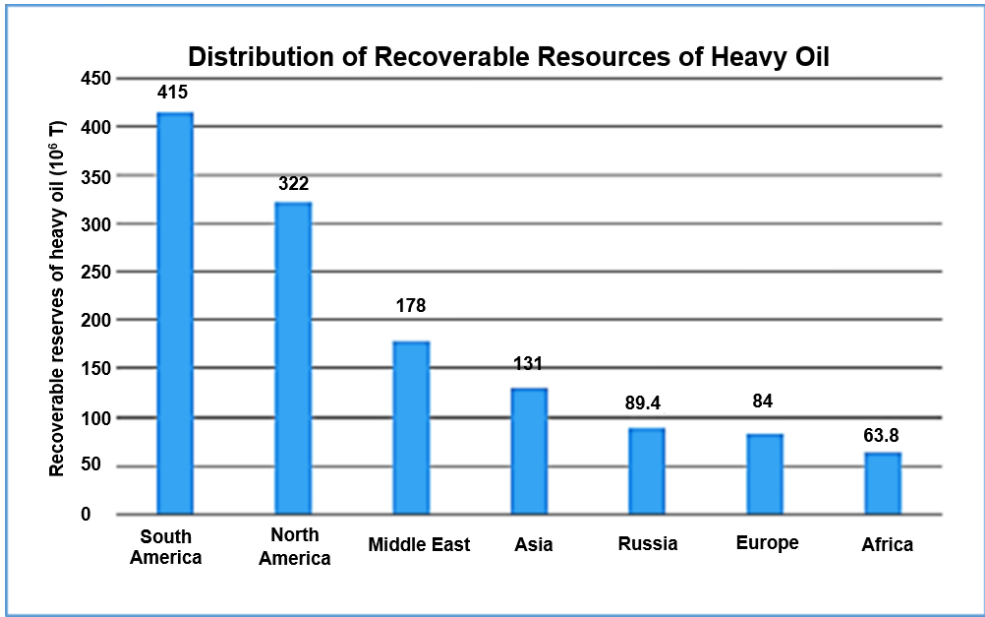


Figure 1. Distribution of recoverable resources of heavy oil in various regions. Adapted with permission from Ref. [6] © 2019 Geological Society of China

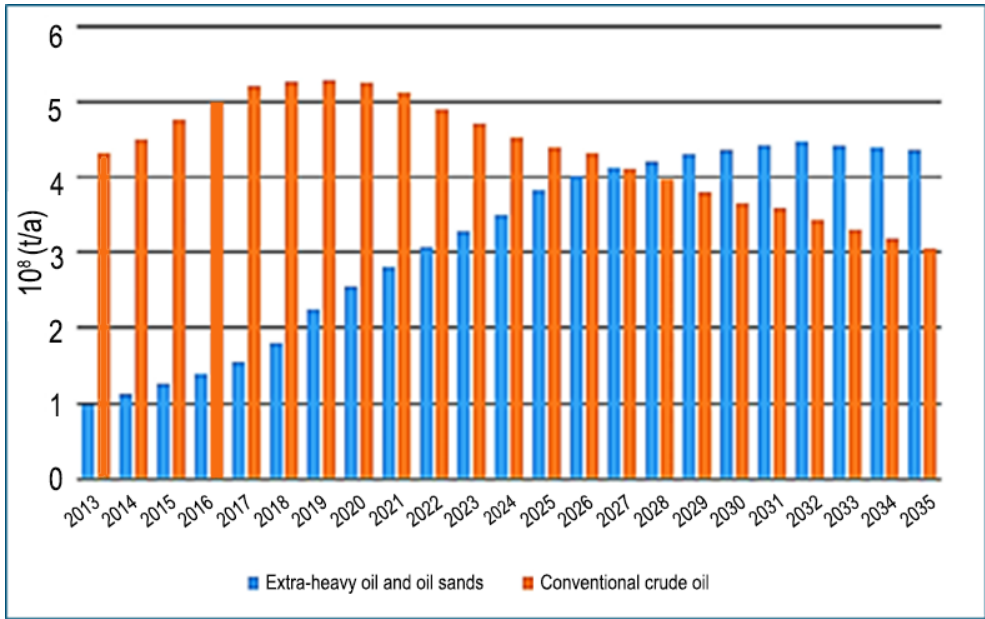


Figure 2. Global heavy-oil production and forecast over the next 20 years. Adapted with permission from Ref. [6] © 2019 Geological Society of China

Drag-reducing agents (DRAs) have been utilised as chemical additives to lower the pressure loss in turbulent flows and to raise crude production. Nevertheless, besides the polymers, surfactants, and fibres that essentially constitute the structural constituents of DRAs, any other solvents or chemical compounds can be also part of DRAs. In fact, a moderate degree of the addition of these additives decreases the crude oil friction, and an increase in the pipeline throughput is attainable without any modification to the pipeline. DRAs can help to reduce the cost of pumping crude oil in pipelines or maximize crude oil flow rate in pipelines [11]. The phenomenon of increasing the rate of drag reduction was first discovered by Toms [12] in 1948. He found that blending a few quantities of a polymethylmethacrylate (PMM) could diminish the drag by as much as forty percent in the amount of drag. The experimental data that was gathered regarding the capacity of polymer solutions to lessen turbulent drag was provided in an article that was a review. The Trans-Alaskan pipeline is the only pipeline that had DRA in it from the time that it was built in 1979. By using the Conoco drag reducer in 10% diluted form, a high-molecular-weight polymer solution in a hydrocarbon solvent, this system was devised.

While many studies have been reported on the role of DRAs in easing the transportation fluidity of light and medium crudes, few reviews addressed heavy and extra-heavy crudes. These unconventional resources pose specific difficulties because of their high viscosity and the high content of asphaltene and resin they contain, which characterizes a marked non-Newtonian behavior with corresponding costly transportation. Previous review works have been either general overalls looking at topics or in a few classes of additives, not bringing together the different experiences, simulations, and green chemistry experiments representing heavy crudes. The objective of the present study lies in closing this gap, providing a comprehensive review of recent advances and limitations in the use of DRAs with a focus on heavy crude oil transportation to sustainably bridge this technology path.

This review also aims to present an extensive knowledge of the current status of research on the use of DRA for delivering heavy crude. Through a summation of available evidence, this paper is intended to provide direction regarding the most fruitful trends in the area and the gaps that need to be addressed in future research. The structure of this review is organized as follows: Section 2 discusses the rheological characteristics of heavy crude oil; Section 3 presents pipeline transportation technologies; Section 4 explains drag reduction mechanisms and agents; Section 5 reviews green technologies; Section 6 outlines the scope of future research; and Section 7 concludes the paper.

2 Rheological Characteristics of Heavy Crude Oil

Crude oil remains a fundamental source of the world's energy supply. Wells are dropping through the production of lighter crude and moderate grade crude due to the increased demand for fuel energy and increased population. It is possible to make use of heavy crude oil as a source of fuel energy in order to protect light crude oil wells from being depleted at an accelerated rate [13]. Crude oil is typically classified into four categories: Light, Medium, Heavy, and Extra-Heavy, with the classification being determined by the API measurement. For classification purposes, the level of sulphur present in heavy crude oil defines its sweetness or sourness. Heavy crude oil has a high viscosity and poor emulsification, and it adheres closely to a surface, so the specific gravity is high and the API is low [14]. Heavy crude density, API, specific gravity, and viscosity are their physical properties. To illustrate, heavy crude is generally constituted by asphaltene, resin, aromatics, saturate, and water. The chemical characteristics of heavy crude include all of these components. Asphaltene and resin content are the two key components that account for the composition of heavy crude [15]. Recent studies have computed the rheological behavior of heavy crude oils using rheometers (rotational or cone-plate) after conditioning the samples at select temperatures. The method involves applying a shear rate sweep and measuring the resulting shear stress to derive the apparent viscosity ($\mu = \tau/\dot{\gamma}$). Essential parameters such as consistency index (k) and flow behavior index (n) have been produced by fitting models (e.g., power law or Herschel-Bulkley) to the flow curves. Beloglazov et al. [16] measured rheological properties of a number of heavy oils under shear rates up to $\sim 300 \text{ s}^{-1}$ and used power-law modeling to establish k and n values.

2.1 Physical Characterization of Heavy Crude Oil

One of the most essential factors in characterizing and separating heavy crude oil from other types of crude oil is density [14]. The mass-to-volume ratio is known as density [17]. Specific gravity is yet another essential component that plays a significant role in evaluating whether or not heavy crude oil is present and in calculating the API. The term "specific gravity" refers to the ratio of the density of heavy crude oil to the density of water, which is the standard density [18]. According to the specifications, the °API of heavy oils should be somewhere in the range of 5 to 20. The range of what is considered to be the viscosity of this particular kind of oil is between 0.1 to 100 Pa·s [8]. In spite of the fact that denser crudes may on occasion display higher viscosities, the viscosity of a heavy crude is not instantly controlled by the conforming API gravity [19]. Alternately, it is potential to decrease the viscosity by rising the temperature without having an effect on the °API, as shown in Table 1 [7].

When referring to the degree of resistance that heavy crude presents to the flow of oil, the term "viscosity" is used. The viscosity of heavy crude is significantly higher than that of light or medium crude oil [20]. This is because heavy crude contains a bigger quantity of asphaltene and resin. The viscosity means the internal resistance to the

flow of crude oil and is described as a ratio of shear stress (τ) to shear rate (γ). Although Newtonian fluids have constant viscosity, for heavy crude oils, the behavior is often shear-thinning, with the viscosity decreasing as the shear rate increases. This measurement, which is mostly affected by temperature and asphaltene/wax concentration, plays an important role in the assessment of flowability and transportation operations [16, 21].

Table 1. Characteristics of various heavy crudes [7]

Area	Country	°API	Kinematic Viscosity (cSt)	
			25 °C	50 °C
Duri	Indonesia	20.8	779	175
Maya	Mexico	21.2	240	66
Cold lake	Canada	21.2	133	40
Zuata	Venezuela	14.8	4349	305
Boscan	Venezuela	8.9	69543	4190

Note: API = American Petroleum Institute gravity degree.

2.2 Chemical Characterization of Heavy Crude Oil

Heavy crude oil is a complex and highly viscous hydrocarbon resource characterized by its high density, high molecular weight, and elevated content of impurities such as sulfur, nitrogen, heavy metals, and asphaltenes [22]. Heavy crude oils typically consist of four main fractions are saturates, aromatics, resins, and asphaltenes. These components influence the oil's physical and chemical properties, where saturate components represent non-polar alkanes with low reactivity, which contribute to fluidity of crude oil, aromatics contain one or more aromatic rings. Carbon and hydrogen make up the bulk of the oil; hydrogen-to-carbon ratios are often lower in heavy crudes, indicating higher aromaticity. Sulfur is present in concentrations up to several percent by weight. The examination of these distributions can be made easier with the help of techniques like mass spectrometry and gel permeation chromatography (GPC) [23, 24].

Understanding the chemical composition of heavy crude is critical for designing effective upgrading and refining strategies, minimizing environmental impact, enhancing transportation and storage, and predicting behavior in thermal and catalytic processes [25].

In summary, chemical characterization of heavy crude oil involves a combination of elemental analysis, molecular structure determination, and compositional profiling. This comprehensive understanding is vital to managing its challenges and unlocking its economic potential.

3 Pipeline Transportation Technologies

The petroleum industry has significant hurdles in the processing, production, and transportation of heavy crude. The central focus of this difficulty lies in the viscosity of the fluid, which is the crucial factor determining the oil's flow ability during the manufacturing process. In the process of moving from the reservoir to the delivery conditions, the temperature and pressure of the oil experience significant differences. These variances can result in considerable changes in phase behaviour and physicochemical properties, which will ultimately have an effect on the thermophysical properties of the fluid [26]. For heavy crude oils, significant variations in circumstances result in a large range in viscosity, spanning many orders of magnitude. Transformations from Newtonian to non-Newtonian rheological behaviour are examples of this type of transition. On account of this, it is of the utmost importance to conduct a comprehensive analysis of the rheological properties of heavy crudes to ensure that the manufacturing process for these oils is both efficient and environmentally friendly [27].

3.1 Rheological Behavior of Heavy Crude Oil with Thermal Treatment

When it comes to the diminution of the elevated viscosity of dense crude oil and enhancing its flowability in pipe systems, heat treatment is a technique that is regularly utilised. When heat is applied to the pipeline, the viscosity of the oil declines rapidly, which in turn lessens the resistance that the oil has to flow (flow resistance). When the crude oil is warmed up to a temperature that is greater than 40 °C, its rheological features remain practically stable at greater shear rates [28]. Shear rate (γ) in crude oil pipelines is specifically influenced by the flow velocity, pipe diameter, and flow pattern. Under laminar flow, the shear rate is proportional to the velocity gradient over the radius of the tube; whereas under turbulent flow, it additionally depends on velocity fluctuations and roughness of pipe walls. As such, both design conditions (i.e., pipe diameter and flow rate) and fluid characteristics (viscosity, non-Newtonian effect) together dictate the applicable effective shear rates encountered in heavy crude conveyance [16, 29].

The influence that temperature has on the standard flow characteristics of dense crude oil is depicted in Figure 3. More particularly, the viscosity and shear rate are especially discussed in connection to temperature. Throughout

a wide range of shear rates, the image provides a striking illustration of the non-Newtonian phenomenon of shear thinning, which is characterized by a considerable reduce in the obvious viscosity upon increasing temperature [30]. Furthermore, it illustrates that the contrast in viscosity is more obvious at minimal shear rates, in contrast to greater shear rates.

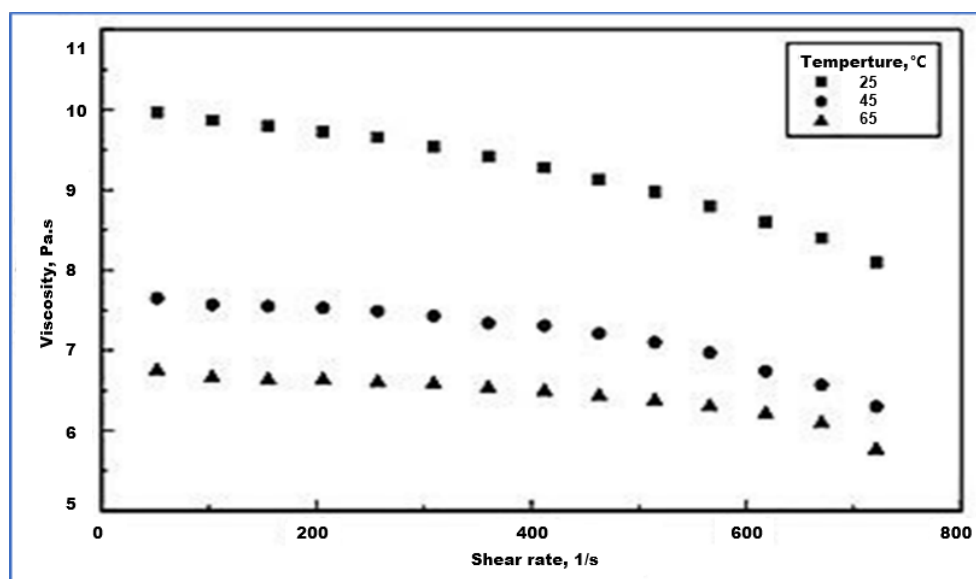


Figure 3. Shear rate-dependent apparent viscosity for heavy crude at various temperatures. Reproduced from Ref. [30] under the Creative Commons CC BY 4.0 license

Heating crude oil requires a lot of energy. High temperatures cause internal corrosion, requiring more heating stations, and heat loss in the pipeline owing to heavy oil flow limitations. Insulating pipelines helps to keep temperatures at a high level and reduces the amount of heat that escapes. It is also possible that rapid pipeline expansion and shrinkage will be troublesome. There is a significant cost involved in transporting heating and pumping equipment over significant distances from the oil field and the storage facility or refinery.

3.2 Rheological Behavior of Heavy Crude Oil with Additives Addition

As a result of the large fall in pressure that occurs when heavy crude is moved through pipelines, particularly over long distances, the amount of pumping energy that is required is going to be increased [31]. A loss of pressure due to friction takes place when the oil flows against the walls of the pipeline, which causes a rise in the amount of energy that is consumed. During pump operation, the fluid pressure in the pipeline is increased to reduce the amount of friction and drag that occurs. On the other hand, this results in an increase in operating expenses and has the potential to overrun the maximum allowable operating pressure (MAOP) of the pipeline [32]. The viscosity of the fluid is related to the pressure loss in the pipeline. At lower pressures, fluids with a higher viscosity drop more. It is essential, risk-free, and straightforward to use additives in order to enhance the viscosity of viscous crude oil in the pipeline. This is because of the different rheological characteristics of the oil.

Gudala et al. [32] investigated the effects of various factors on pressure drop, shear viscosity, pumping power savings, and flow increment during heavy crude flow in horizontal SS304 stainless steel pipelines. These factors included diameter (0.0381 m and 0.0508 m I.D., both 2.5 m long), temperature (25 °C to 50 °C), water cut (0% to 15%), and surfactant *Madhuca longifolia* (0% to 2000 ppm). Adding 2000 ppm of Mahua to 85% heavy oil and 15% water in a 0.0508 m I.D. pipeline at 50 °C and 7.2 meters per hour resulted in a reduction of 130.8% in power consumption and an increase of 121.8% in flow. Therefore, natural surfactant that has been extracted is an excellent emulsifier, viscosity reducer, flow improver, and alternative to commercial surfactant.

For the objective of determining the rheology of three samples of crude oil from Kuwaiti wells, Al-Adwani and Al-Mulla [33] conducted tests. An instrument known as a controlled stress and strain rheometer was used to measure this. A 14.81 API is found in crude A, 16.33 API in crude B, and 14.09 API in crude C. The researchers investigated the effects of polyacrylamide (PAM) concentrations of 40, 50, 70, and 100 ppm on rheology. In addition, they tested PAM to five different surfactants, which were as follows: cetyl-trimethylammonium bromide (CTAB), sodium-dodecyl sulphate (SDS), poly-(sodium 4-styrenesulfonate) (PSSS), hexa-decylphosphocholine (HTPC), and CHP, which was a mixture of PSSS, CTAB, and HTPC in equal molar ratios. Figure 4 illustrates that the viscosity of processed crude samples was lowered when 70 ppm of PAM was added to them in comparison to pure samples.

At a surfactant concentration of 70 ppm, PSSS and CHP had the lowest viscosity for Crude A and B, respectively. Loss modulus (G'') was raised by either PSSS or CHP.

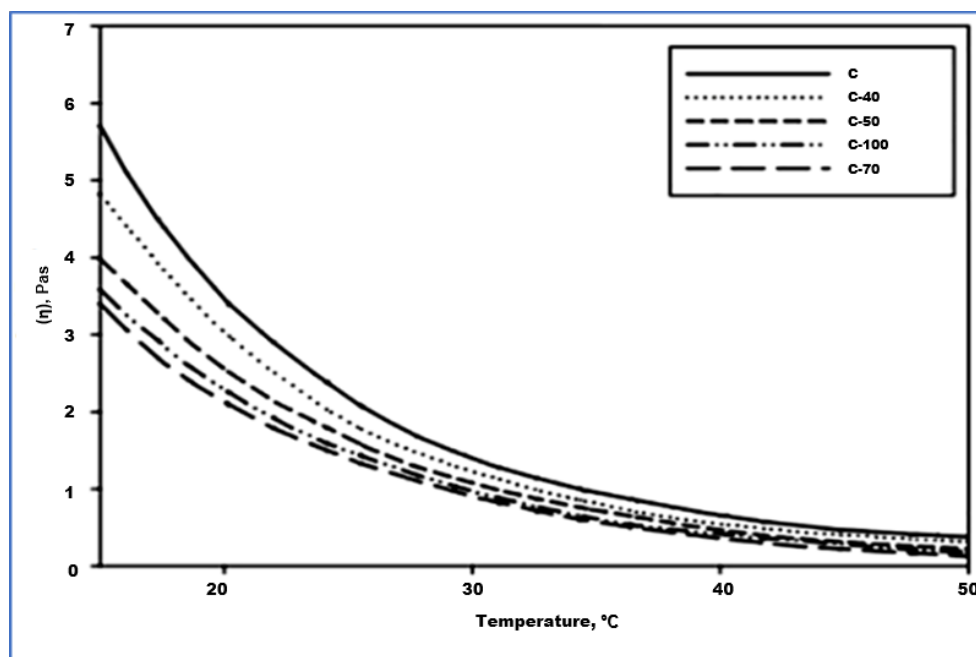


Figure 4. The viscosity characteristics of Kuwaiti heavy crude to temperature and polyacrylamide (PAM) content. Reproduced from Ref. [33] under the Creative Commons CC BY 4.0 license

Chitosan-based cationic surfactant (CBCS) was synthesized by Negi et al. [34]. This surfactant is more eco-friendly. When such a surfactant was employed, the viscosity of a heavy crude could be decreased. The chemical structure of the surfactant was characterized by IR, NMR, TG, and XRD. This was carried out to investigate the effect of synthetic surfactant on heavy crude viscosity. From 0 to 600 ppm, surfactant was observed to increase the crude viscosity, their study found.

3.3 Rheological Behavior of Heavy Crude Oil in Water Emulsion

Emulsions are made up of a dispersal of a liquid that does not mix well with another liquid, which is referred to as the dispersed phase, and a different liquid, which is referred to as the continuous phase [35]. Multiple emulsions are made up of droplets of a continuous phase that are surrounded by droplets of a dispersed phase. One example of this is the water-in-oil-in-water (W/O/W) emulsion. It is possible to considerably reduce the risk of wax deposits on the pipe surface, pipe clogging, and pipeline corrosion by ensuring that water is the continuous phase throughout the process. The optimisation of the operational efficiency of the transportation process is always a top priority for the oil sector. It is essential to lessen the viscosity of crude while simultaneously increasing the amount of oil it contains in order to reach the highest possible levels of efficiency and cost-effectiveness [36].

3.4 The Rheological Behavior of Heavy Crude Oil Varies with Dilution

Dilution has been a procedure that has been utilized extensively since the 1930s to minimize the high viscosity of heavy crude. When a crude oil with a high viscosity is mixed with a less viscous fluid, the viscosity of the crude oil is reduced to a level that is adequate for pumping. This is the basic concept behind this approach. Dilution is found to be an effective method in enhancing the flow properties of crude oil and facilitating certain processes (e.g., dehydration, desalination) [37, 38]. In order to do so, two separate pipelines must be used: a production line for the oil itself and a separate diluent pipeline. A solvent ratio of crude oil from 20% to 30% is enough to improve the crude oil transportation by pipeline [4]. To decrease the viscosity of heavy crude to an extent suitable for transportation, dilution is performed with a very large amount of diluent, as great as 30% of the total used. Because of this, there is a rise in the quantity of oil that is being transported, which in turn necessitates a significant expansion in the capacity of the pipeline. Consequently, significant financial resources are necessary for the construction of pipelines, leading to an escalation in costs [39]. This leads to the precipitation of the asphaltenes in dilute phases, which results in unstable transportation or storage. This instability can lead to pipeline plugging, necessitating the testing of the oils for their compatibility. Moreover, because of lower quality, the heavy crude blend price also becomes less than the diluent price [37]. The dilution approach, allowing for addressing the chemical composition, density, and rheology

of the heavy crude, was rather effective and achieved a significant quality enhancement of the crude as the main direction of improvement [40, 41]. The diluent oils that were commonly used with the heavy oils were propane, toluene, heptane, naphtha, heavy oil distillation fractions, and light crude oil. As an additional point of interest, Ilyin and colleagues suggest the use of bio-oil as a diluent oil. Bio-oil is produced by the thermochemical processing of wastes from agricultural and forest sources. Because of the significant processing that these highly refined oils go through, they are not cost-effective [42]. The demulsification and viscosity reduction capabilities of light crude oil are significantly less than those of heavier crude oil [43]. In an effort to lessen the viscosity behavior, a study has been carried out to evaluate the impact that light oil concentration has on the behavior of viscosity. Ibrahim et al. [44] evaluated the impact of dilution on the viscosity of heavy crude oil. This was done in response to earlier studies that revealed the electric field can reduce viscosity. One of the diluents that was utilized was acetone at a variety of concentrations. In an effort to lessen the viscosity of Iraqi crude oil, Azeez et al. [45] utilized a mixture of toluene and dimethyl ketone as diluents in varying proportions (50/50 vol.%).

4 Drag Reducing

The term “drag reduction (DR)” refers to a phenomenon in the field of transportation in which the incorporation of a modest quantity of material can contribute to a reduction in the friction factor of fluid flow [46, 47]. The fundamental objective of the function is to improve the effectiveness of oil pumping by the utilization of active agents that are referred to as DRAs [48, 49]. Consequently, this leads to an increase in capacity as well as a decrease in the amount of power that is necessary for the pumping process. Drag reduction, expressed as the percentage decrease in drag (%DR), is calculated by comparing the pressure drop value (Δp_0) without an additive to the pressure drop value (Δp_{DRA}) with an additive [44, 49, 50].

$$DR\% = \left(\frac{\Delta p_0 - \Delta p_{DRA}}{\Delta p_0} \right) \times 100 \quad (1)$$

where,

Δp_0 : pressure drop value without DRA.

Δp_{DRA} : pressure drop value with DRA.

4.1 Drag Reducing Agents

Flow efficiency is a critical challenge. One of the most effective techniques to address these issues is the application of DRAs. These are chemicals, such as specific types of polymers, surfactants, or very small additives, that are added to the fluid in very small volumes in order to reduce the friction that is created by turbulence in pipes [46]. It is possible to observe the influence of drag reduction by analysing the morphology of the component ions and the macromolecular architecture of every variation of the surfactant [47]. The incorporation of DRA (in ppm) has the potential to optimize the cost benefits associated with transporting diluted crude. Turbulence is reduced as a result of its ability to inhibit the formation of turbulent eddies, which in turn enables bigger flow rates to be achieved while maintaining the same pressure drop [7].

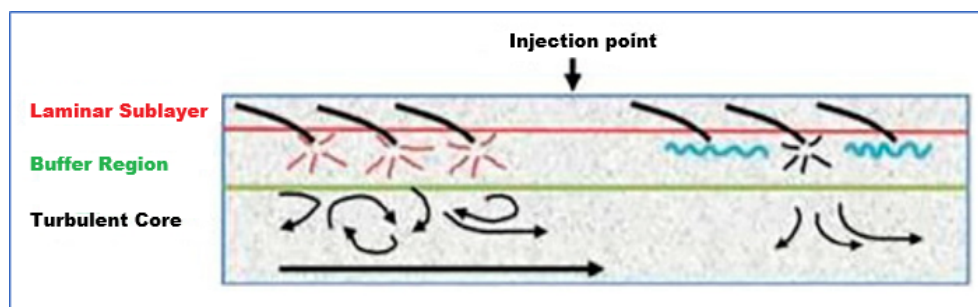


Figure 5. Schematic representation of turbulent flow within the pipeline. Adapted from Ref. [48] under the Creative Commons CC BY 4.0 license

Fibres and polymers often reduce friction thereby decreasing turbulent eddies. The flow transition from the laminar zone to the turbulent layer is disrupted as a result of the total expansion of polymer molecules, as depicted in Figure 5 [48]. DRA were implemented in the Trans-Alaska Pipeline System, which resulted in a thirty percent increase in the flowability of crude oil. Other maneuvers, such as water disposal and firefighting, have also been carried out with the assistance of DRAs [47]. Nanomaterials show great promise in enhancing pipeline transport efficiency by reducing drag. Ongoing research focuses on optimizing formulations, ensuring environmental safety, and scaling up for field applications [51, 52]. Several studies have explored the effectiveness of nanomaterials, particularly

Nano-Silica, in reducing drag in Iraqi crude oils [53–56]. These studies demonstrate significant improvements in flow properties and viscosity reduction, contributing to enhanced pipeline transportation efficiency.

Table 2 offers a consolidated overview of previous studies on the rheological behavior of heavy crude oils with additive additions, nanomaterials applied to crude oil pipelines, and Iraqi crude oil case studies by highlighting key experimental parameters, additives, and their main outcomes, and a more consistent and comprehensive understanding of drag reduction mechanisms across different studies is achieved.

Table 2. Integrated summary of experimental studies on rheological modification and drag reduction of crude oils with additives and nanomaterials (including Iraqi case studies)

Ref.	Crude Type	Additive/Nanomaterial	Experimental Setup/Conditions	Key Findings
Gudala et al. [32]	Heavy crude	Natural surfactant (Madhuca longifolia)	Varied diameter, temperature, water cut	Good emulsifier; viscosity reduction.
Al-Dawery and Al-Shereigi [31]	Heavy crude	Palm fiber, walnut shell, roasted date kernel	Rheological tests	Viscosity dropped by 15 %; bio-materials reduced yield stress
Al-Adyani and Al-Mulla [33]	Crude A & B	Polyacrylamide (PAM), PSSS, CHP	70 ppm surfactant; rheometer	Lowest viscosity achieved.
Negi et al. [34]	Heavy crude	Chitosan-based cationic surfactant (CBCS)	0-600 ppm; rheological tests	Viscosity increased with CBCS
Alsaedi et al. [50]	Crude oil pipelines	TiO ₂ and Cu nanoparticles	Flow enhancement using nanoparticle agents	Drag reduction up to 78%
Wu et al. [51]	Crude oil	Carbon dot-based nanofluids	Stable, salt-tolerant nanofluids	Demonstrated drag reduction
Esfandiari et al. [52]	Crude oil pipelines	Functionalized magnetic TiO ₂ + Lecithin	Different pipeline materials	High drag reduction 85.2–99.1%
Azeez and Hussien [53]	East Baghdad heavy crude	Nano-Silica + Ultrasound	1500 mg/L at 35°C	Ultrasound-enhanced performance 65 % drag reduction;
Ibrahim et al. [54]	Iraqi crude oil	PMMA Polymer + CNTs	3000 ppm	significant improvement in pipeline flow
Shakir and Al-Tameemi [55]	Iraqi crude oil	Nano-CuO	Moderate concentrations	Up to 4% viscosity reduction

Note: PSSS = poly-(sodium 4-styrenesulfonate); CHP = an equimolar mixture of cetyl- trimethylammonium bromide (CTAB), hexa-decylphosphocholine (HTPC), and poly-(sodium 4-styrenesulfonate) (PSSS); PMMA = polymethyl methacrylate; CNTs = carbon nanotubes.

4.2 Mechanism of Drag Reduction

The operational mechanism of DRA additives can be categorized as follows: (1) the polymer serves as a DRA by mitigating the formation of eddies, thereby lowering turbulence. Additionally, it has been noted that polymer performance is influenced by secondary parameters, such as concentration, pipe size and shape, molecular weight, chain flexibility, and flow velocity [55]. (2) Fibers are used to reduce drag without causing environmental contamination issues, as they are non-reactive with substances and do not generate harmful compounds. These structures can alter the transfer properties of a fluid solution. Dong et al. [56] investigated a novel superhydrophobic fibrous network constructed from cross-shaped polyester fibers coated with polypyrrole. Simultaneously, the unique geometry of the base fibers triggered the Concus-Finn (CFin) capillary effect, promoting rapid oil transport through the network. (3) Surfactant additives decrease the resistance inside the pipeline by creating micelles through the interaction among polar and nonpolar molecules of the surfactant and crude. These micelles act as impact absorbers, reducing flow turbulence and minimizing the formation of eddies [49]. A rheological test and a drag reduction simulation device were utilised by Yuan et al. [57] in order to investigate the possibility of adopting additive drag reduction technology in water injection pipelines. According to the results of the tests, the selected cetyltrimethylammonium chloride (CTAC) exhibits a high degree of thixotropy, with the viscosity recovery rate nearing 97% in a time span of less than 300 seconds. Furthermore, the CTAC/NaSal (sodium salicylate) combination offers many benefits, including the ability to tolerate both salt and oil. (4) Nanofluids consist of tiny particles measured in nanometers. The particles collide within a fluid medium in the form of a suspension [58]. Nanofluids have distinct drag reduction properties compared to surfactants and polymers. The use of the surface modification approach with nanofluids reduces drag within a pipe. Nanoparticles deposit in the cracks along the inner wall of the pipe, resulting in smoother liquid flow through the pipeline. A smoother wall leads to reduced turbulence in the flow, resulting in a decrease in the Reynolds Number (Re) [59]. This benefit can be attributed to the reduction in wall roughness inside

the pipe. Nano-Silica is the most suitable nano-fluid for reducing drag. It is pretty inexpensive and straightforward to make. It has been demonstrated to be an effective drag reduction when added to crude oil and other fluids, as illustrated in Table 3.

Table 3. The operational mechanism of drag-reduction agent (DRA) additives

Ref.	Focus	Summary
Ibrahim et al. [54]	Polymer	The polymer reduces drag by minimizing eddies and turbulence.
Dong et al. [56]	Fiber	The ability to alter the transfer properties of a fluid solution.
Yuan et al. [57]	Surfactant	Surfactant additives reduce pipeline resistance by forming micelles from polar and nonpolar surfactant and oil molecules.
Alhamd et al. [59]	Nano fluids	Nanoparticles are deposited in the cracks along the inner wall of the pipe, resulting in a smoother flow of liquid through the pipeline.

4.3 Application of Drag-Reducing Agents in the Pipeline Transportation of Heavy Crudes

Because of the simultaneous rise in the demand for energy on a worldwide scale and the slowdown in the production of conventional crude, heavy crudes are frequently considered to be the energy resource of the future. Pipelines are often used to transfer heavy crude oil from production sites to storage facilities or refineries located in a variety of places across the globe. The viscosity of crude oil is the most substantial physical feature that determines how it flows through pipelines. Because it generates shear or frictional forces through the Constituent particles of the fluid and the walls of the boundary, viscosity is a factor that makes movement more difficult [39]. To combat the growing shear and friction forces, the high viscosity of crude necessitates the utilization of a substantial amount of pumping power in pipelines traversing long distances. As a result, it is essential to lessen the viscosity of these thick oils in order to simplify the process of pumping them, reduce the expenses of operations, and lessen the negative impacts of pressure dips in pipelines that occur over transportation and processing [60]. In terms of transporting heavy crude in pipelines over a long distance, the economic benefits of applying DRA to diluted heavy oil can be positive. a few researchers have investigated in this field, on the other hand. Earlier study mainly focused on the application of DRA in transporting low density crude oils.

Rashid et al. [61] studied the influence of polyacrylic acid concentration as a polymer on pressure losing and drag reduction in the flowing Iraqi heavy crude in a tow tube. It was observed that treatment with 250 ppm of PAA resulted in 16% drag reduction and 29% pumping power reduction. The introduction of DRA was found to increase flow rate by 4%. In 2019, Al-Adwani and Al-Mulla [33] conducted research that focused on the rheology behavior of crude oil samples extracted from Kuwait. The researchers investigated how these qualities are influenced by variables such as the gravity of the API, the concentration of the drag reduction material, the isothermal and non-isothermal circumstances, and the speed and frequency of the mixer. There were three different kinds of heavy oils from Kuwait that were subjected to an analysis. The API values for these oils were 14.8, 16.3, and 14.1. Evaluation was performed to determine the drag reduction that was brought about by the introduction of a number of different DRAs. In the experiment, five different surfactants were used as DRAs. These surfactants were as follows: i. SDS, ii. HTPC, iii. PSSS, iv. CTAB and v. CHP, which was a mixture of PSSS and CTAB in equal molar fractions. According to the findings of the studies, the maximum drag reduction was achieved by utilising CHP and PSSS at a concentration of 70 ppm by volume.

4.4 Solid Particles as Drag-Reducing Agents

Solid suspensions are classified as insoluble additives and can be broken down into two categories: fibrous suspension and non-fibrous suspension [62, 63]. As a result of the fact that their basic components are derived from natural resources, these suspensions are economically viable [64]. The research that Hashizume [65] conducted focused on the ability of three distinct types of fibrous suspensions to reduce the amount of drag that they experienced. They discovered that the maximum drag reduction for rayon is 18%, while the maximum drag reduction for nylon is 8%, and the maximum drag reduction for asbestos is 70%. In light of the fact that synthetic fibers and asbestos both contain synthetic compounds, researchers have recently begun investigating fiber suspensions, which are derived from natural resources. If they are drained without being properly treated, these synthetic compounds have the potential to damage the environment [66]. Shenoy [67] conducted earlier research that generated results that are congruent with these findings. In that study, asbestos suspensions that were blended with surfactant caused a reduction in drag of up to 44%. After doing their research, Xi [68] came to the conclusion that the combination of both types of additives results in a decrease of over 95% in drag. This finding suggests that the two types of additives may react variously with turbulent flow. Pamitran et al. [69] propose the use of fiber suspensions to reduce the amount of drag that is experienced. Initially, at low velocities, fibers accumulate near the pipe wall, forming a dense network that

increases friction and results in no drag reduction (Figure 6a). As the flow velocity increases, a fluid annulus forms between the fiber network and the pipe wall (Figure 6b), resulting in a slight reduction in shear and initiating minimal drag reduction. With further velocity increase, the fibers align more parallel to the flow, allowing a water layer to form along the pipe wall (Figure 6c). This alignment minimizes wall roughness and shear, achieving maximum drag reduction. Indeed, recent experimental studies have shown promising synergistic drag-reduction effects for hybrid systems, for example, combining polymers with nanoparticles (e.g. Qin et al. [70]), which may outperform the use of a single additive.

Nano-Silica is the most suitable nano-fluid for reducing drag. It is quite inexpensive and straightforward to make. It has been demonstrated to be an effective drag reduction when added to crude oil and other fluids.

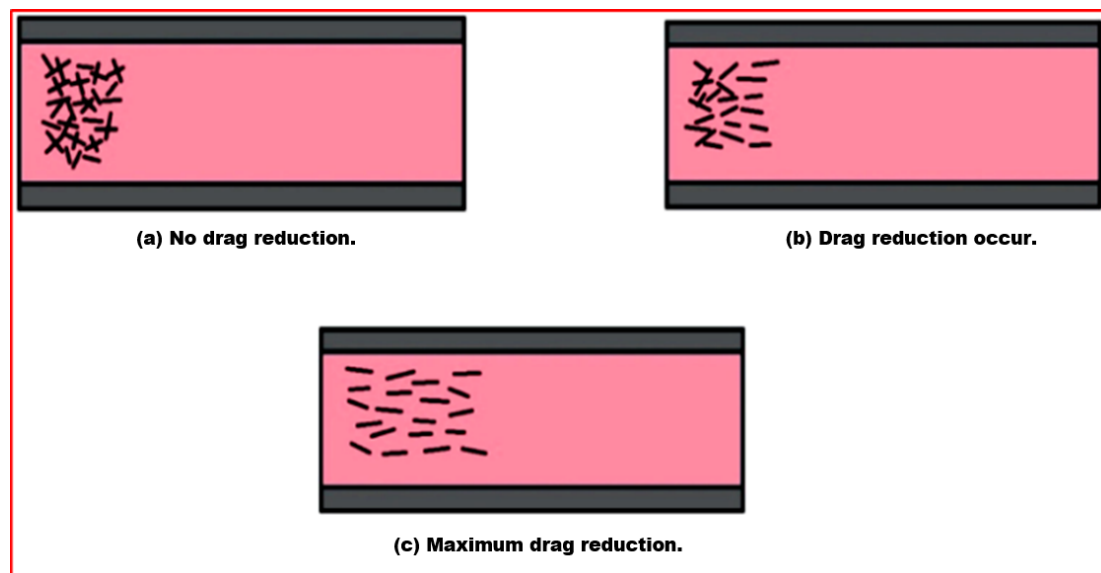


Figure 6. Mechanism of drag reduction of fiber suspensions. Adapted from Ref. [69] under the Creative Commons CC BY 4.0 license

4.5 Nano-Particles as Drag-Reducing Agents

Nanoparticles (NPs) are diminutive particles, ranging from 1 to 100 nanometers, possessing remarkable penetrating and adsorption capabilities, adjustable physicochemical properties, and distinctive thermal characteristics [71]. Moreover, due to their diminutive dimensions, nanoparticles can traverse minuscule apertures and constricted passages that are inaccessible to bigger materials.

Over the course of the last few decades, nanotechnology has rapidly developed as a revolutionary and dominant approach that is capable of competing with older technologies in terms of both their technological and economic capabilities. Its applications in the oil and gas industry offer potential that have never been seen before for the development of oil and gas extraction methods that are more cost-effective, efficient, and environmentally friendly [72, 73].

Nanotechnology has been demonstrated to be an efficient method for enhancing the flow characteristics of heavy crude oil in pipelines by lowering the viscosity of the oil to be transported. The higher surface-to-volume ratio of nanomaterials allows these materials to offer various advantages in the solvent desalting process. These benefits include having many active sites available, which help attract asphaltenes more effectively, resulting in better removal of asphaltenes [74]. The main and frequently used component in inorganic nanoparticles is silica [75]. According to Tajabadi et al. [76], Silica (SiO_2) nanoparticles are a promising enhanced oil recovery (EOR) agent in reservoirs with water-wet sandstone (SiO_2) particles as an EOR agent. It is observed that SiO_2 particles show amazing thermal stability; it was found that even at 65°C treatment temperature, their specific surface area hardly changes, as reported for the study [77, 78]. This is evidenced by the fact that SiO_2 particles have been shown to have outstanding thermal stability.

The incorporation of silanol (Si-OH) groups onto the surface of hydrophobic silicon oxide nanoparticles has been shown to yield favorable results, indicating that these modified silica nanoparticles serve as superior EOR agents in sandstone reservoirs compared to metal oxide nanoparticles [79]. Table 4 provides a comprehensive summary of the most common types of nanoparticles used in the oil, chemical, and cement industries. This classification indicates which type of nanoparticle is most suitable for industrial applications, such as reducing drag, enhancing oil recovery, altering viscosity, and strengthening cement.

Table 4. Summary of nanoparticles and their applications

Nanoparticle Type	Formula/Examples	Specifications	Applications & Advantages	Ref.
Polymeric nanoparticles	PLA (C ₃ H ₄ O ₂) _n , PAA (C ₃ H ₄ O ₂) _n	Water-soluble polymers, high viscosity, shear-sensitive	Reduce drag in crude oil pipelines; flow improvement	[80, 81]
Magnetic nanoparticles	Fe ₃ O ₄	Superparamagnetic; oil-wettable surface; <100 nm	Reduce viscosity of heavy oil; magnetically controlled enhanced oil recovery (EOR)	[82]
Mesoporous silica nanoparticles	SiO ₂	High surface area; porous; chemically stable in brine	Strengthen cement; thermal stability; EOR injection aid	[83]
Silver nanoparticles	Ag, AgNO ₃	Antimicrobial; conductive; easily dispersed	Antibacterial cement additives; corrosion protection	[84]
Chitosan nanoparticles	(C ₆ H ₁₁ NO ₄) _n	Biopolymer-based; biodegradable; pH sensitive	Improves wettability; corrosion control; EOR and cement	[85, 86]

4.5.1 Classification of Nano-Silica particles

The element silicon (Si) has the atomic number 14 and the atomic mass of 28.086. It is a metallic, shiny silvery grey colour. After oxygen, silicon is the second most abundant element in the lithosphere, accounting for 27.5% of the total composition of the lithosphere. It is only in the forms of oxides and silicates that it can be found in its elemental representation [87]. The typical starting ingredients consist of chunks of quartzite and coke coal, in addition to charcoal and wood chips, which are used to ensure that the charge is ventilated effectively. Silicones have been used directly or indirectly in many industries. The silicon element, with a high degree of purity, holds special importance for the global economy [88]. Based on an estimate, it was determined that Iraq, the Arab World, and the Middle East import silicones with a value of several hundred million dollars. Consequently, silicon and silicone technology are crucial for the development of Iraq's national economy [89]. This is because Iraq possesses significant amounts of high-purity silica sand (SiO₂ > 98%), as well as a presence in the oil and petrochemical sectors. Nanoparticles of silica are utilized in the manufacturing process of a variety of products, such as thermal insulators, humidity sensors, electronic substrates, and electrical insulators. In the petroleum industry, silica nanoparticles are poised to bring about significant benefits with their application. The most common uses to date include the decrease of water invasion in shale, the filtration and rheological management of fluids, the cementing of oil wells, the stability of foam and emulsions for better oil recovery, and the reduction of drag in porous media [90].

4.5.2 Preparation of Nano-Silica particles from local materials

SiO₂ has been the subject of considerable research due to its numerous benefits for various applications. In addition to being a natural component of sand, silica can also be found in a wide variety of other materials, such as rice husk, coffee husk, wheat husk, sugarcane bagasse, corn cob ash, and fly ash. There is a remarkable abundance of silica all across the world, which accounts for 59 percent of the crust of the Earth [91]. A significant amount of silica sand deposits can be found in the Western Desert of Iraq [92]. Nano-Silica has become increasingly popular as a result of its low toxicity, high surface area that is highly reactive, and resilient properties in both the physical and chemical realms [93]. In addition, silicon nanoparticles (SNPs) have a significantly large surface area despite having a relatively tiny diameter, which enables them to be utilized in a wide range of applications. Through the utilization of nano-silica, they further enhance the solvent deasphalting process, which has a significant impact on the quantities of deasphalted oil and pitch that are generated [94].

Nano-silica was synthesized through several methodologies, including chemical precipitation, sol-gel technique, and high-temperature vaporization. Among the methods that have been employed to synthesize silica nanoparticles are the reverse micro-emulsion method, flame synthesis, and the commonly used sol-gel process [95]. The sol-gel process is frequently used for the preparation of silica, glass, and ceramic materials [96]. This is because it can yield pure products that have some homogeneity under manageable conditions. Catalyzed by a mineral acid such as HCl or a base such as NH₃, the process consists of hydrolysis and condensation of metal alkoxides (Si(OR)₄), like tetraethyl orthosilicate (TEOS, Si(OC₂H₅)₄) or inorganic salts such as sodium silicate (Na₂SiO₃). This technique is a method that has been used for a long time. The sol-gel method for making silica using silicon alkoxides (Si(OR)₄)

is shown in Figure 7 [97].

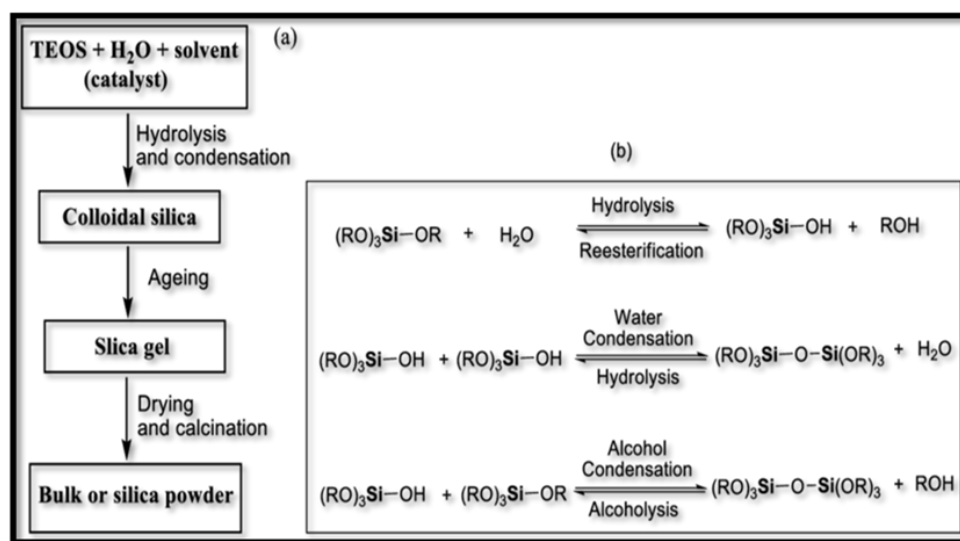


Figure 7. A typical sol-gel process flow schematic. Adapted from Ref. [97] under the Creative Commons CC BY 4.0 license

4.5.3 Advantage of Nano-Silica as drag-reducing agents

DRA substantially influence shale hydraulic fracturing. Therefore, it is imperative to define superior design requirements for engineering applications [98]. Over the course of the past few years, a large number of researchers from both the academic and industrial sectors have used the integration of a small quantity of inorganic nanoparticles into polymers in order to improve the desirable features of these materials [99]. The nanoscale dimensions of nano-silica, the large number of hydrogen bonds it has, and the increased specific surface area it possesses are the primary reasons for its research [100]. In their study, Singh et al. [101] found that the silica fluid has a favorable influence on the reduction of mechanical deterioration. As a result of their increased surface free energy and nanoscale size, the original silica nanoparticles have a tendency to aggregate into bigger particles. Given these circumstances, inhomogeneous dispersion within a polymer matrix is likely to occur, which will have a negative impact on the performance that was planned. To upgrade the similarity of silica nanoparticles with an organic polymer matrix, the surface of the silica needs to be modified [102]. For the purpose of modifying the initial silica nanoparticles, a silane coupling agent is frequently utilized by certain researchers. Silica nanoparticles that have been modified exhibit improved adherence with organic polymers and a very regular dispersion [81].

5 Application of Green Technology in the Transportation of Heavy Crudes

According to what was mentioned earlier, several different tactics are being utilized to improve the conveyance of heavy and extra-heavy crude through pipeline networks, which will ultimately allow for the fulfillment of the growing energy demand [102]. It is possible for some of these processes, such as combining naphtha with an organic solvent, to result in the production of hazardous gases that are responsible for a variety of environmental hazards [103]. Therefore, in order to transport heavy and extra-heavy crude oils, it is necessary to build “green” systems that are not only technically practicable but also environmentally sustainable [104]. This is because the utilization of nanofluids may promote the utilization of naphtha for transportation purposes. Several laboratory tests showed that using nanoparticles and electric or magnetic treatment, heavy oil viscosity might be decreased [105]. The paraffins or asphaltenes tend to agglomerate under a magnetic or an electric field, which lowers the oil’s viscosity. Nanofluid, as a specific engineered mixture, is known to lower the interfacial tension and to modify the flow behavior of crude emulsion. The goal is to develop the rheology of heavy crude. Through a combination of an electrical field and Nano-Silica additives, it can also be used to simplify the flow of heavy crude by positioning paraffin and asphaltene particles as short chains [106]. This results in a lesser viscosity material, and the fluid layers can more readily slide upon one and the other with lesser friction, allowing for easier passage of oil through the layers [107].

6 Scope of Future Research

The competitive behavior between inertial eddy and viscous force can be found when the drag and the viscosity are reduced simultaneously. There is a prevalent association between DRAs and ionic surfactants, which are composed of polymers that are susceptible to breakdown when subjected to shear pressures [7]. Finding the best amount of

the DRA and the appropriate polymer molecular size is crucial to minimize potential damage [108]. In addition, it is essential to conduct an exhaustive investigation into the length of the pipeline in order to determine the degree of reduction in drag at particular Reynolds numbers that correspond to various turbulent regions [65]. It is necessary to acquire simulation tools that have been validated by using experimental data on drag reduction for Reynolds number variation [109]. This is necessary in order to accurately foresee the drag behaviour at optimum conditions of polymer concentration, surfactant polymer kind, oil viscosity, and surfactant solubility [110]. To confirm the prevalent adoption of DRA at the transportation of unfamiliar heavy crudes through pipelines, extensive studies are required. The next paragraphs provide an overview of the essential tasks that will need to be completed in the future to financially deploy this flow technology.

7 Conclusions

This article evaluates previous advancements in methodologies designed to facilitate the long-distance transportation pipelines of heavy crudes. Techniques such as emulsification, dilution, heating, chemical additive fluids, nanoparticles, and DRAs have been explored to moderate the challenges posed by the crude oils with high viscosity. Nanoparticles are a promising agent of DRAs due to their unique physicochemical properties, like high surface area, more stability under severe conditions, and the ability to adjust flow characteristics at the molecular level.

An overall resource on the viscous crude oils is created by compiling data from more than 155 publications. This review's thorough examination of the use of different DRAs and the ways in which they modify the rheological characteristics of heavy oil under various processing circumstances is one of its most noteworthy features. Recent studies show that nanoparticles, especially oxides of metals (e.g., SiO_4 , Al_3O_3 , TiO_2) and carbon nanomaterials, can significantly reduce frictional pressure losses by changing the viscosity profile and furthering laminar flow regions within turbulent regimes. Moreover, nanoparticles demonstrate superior stability under high temperature and shear conditions compared to traditional DRAs. They can help reduce common operational issues such as deposition of wax, asphaltene, and pipeline fouling, leading to improved flow and decreasing maintenance periods in transportation pipelines. Importantly, nanoparticles can be engineered to be environment-friendly, degradable, or can be recovered, supporting the development of sustainable and eco-friendly pipeline solutions. Furthermore, optimizing nanoparticle classifications, concentrations, dispersion mechanisms, and harmony with crude oil compositions through managing comprehensive studies to assess the ecological outcomes of DRA usage and degradation products, and recognizing optimal factors for DRAs application to achieve higher efficiency and minimal environmental impact.

Finally, the development of environmentally sustainable and effective technologies for the conveyance of high-viscosity and extra-heavy crude oils can be undertaken, balancing technological feasibility with ecological reliability. The integration of nanotechnology into heavy crude oil transportation offers a transformative opportunity to upgrade flow efficiency, reduce energy waste, and support environmental sustainability. Continued interdisciplinary research and collaboration between academia and industry will be crucial for realizing the full potential of nanoparticle-enhanced drag reduction in crude oil pipeline systems.

Author Contributions

Conceptualization, S.W.A. and T.J.M.; formal analysis, S.W.A.; investigation, S.W.A.; resources, T.J.M., A.M.A.K., and M.A.A.-B.; data curation, S.W.A.; writing—original draft preparation, S.W.A.; writing—review and editing, T.J.M., A.M.A.K., and M.A.A.-B.; visualization, S.W.A.; supervision, T.J.M.; project administration, T.J.M. All authors have read and agreed to the published version of the manuscript.

Data Availability

No new data were created or analyzed in this study. Data sharing is not applicable to this article.

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Conflicts of Interest

The authors declare no conflicts of interest.

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